

VELLORE INSTITUTE OF TECHNOLOGY

CHENNAI CAMPUS

DATA SCIENCE PROJECT

FINAL REVIEW REPORT

WINTER SEMESTER 2025–26



***"From Data to Success:
Code-Forces Analyzer for Performance Enhancement"***

Field	Details
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SECTION 1

Abstract & Introduction

1. ABSTRACT

COMPETITIVE PROGRAMMING PLATFORMS SUCH AS CODEFORCES GENERATE VAST AMOUNTS OF BEHAVIORAL DATA RELATED TO USER PERFORMANCE, CONTEST PARTICIPATION, AND PROBLEM-SOLVING PATTERNS. HOWEVER, THIS RAW DATA IS RARELY TRANSFORMED INTO ACTIONABLE INTELLIGENCE THAT HELPS USERS IMPROVE SYSTEMATICALLY. THIS PROJECT, TITLED CODEFORCES AI COACH, ADDRESSES THIS GAP BY DEVELOPING A MACHINE LEARNING-POWERED SYSTEM THAT ANALYZES USER BEHAVIOR AND PREDICTS SHORT-TERM RATING CHANGES ON THE CODEFORCES PLATFORM.

THE SYSTEM INTEGRATES PRACTICE BEHAVIOR, PROBLEM DIFFICULTY PATTERNS, AND CONTEST PERFORMANCE TRENDS TO PROVIDE BOTH PREDICTIVE ANALYTICS AND PERSONALIZED COACHING RECOMMENDATIONS. BUILT ON A DATASET OF 7,029 USER SAMPLES AND POWERED BY A GRADIENT-BOOSTED XGBOOST MODEL, THE APPLICATION DELIVERS MONTHLY GAIN PREDICTIONS, WEAK TOPIC DETECTION, 12-MONTH GROWTH SIMULATION, AND A TAILORED PRACTICE PLAN — ALL THROUGH A STREAMLIT-BASED INTERACTIVE WEB INTERFACE.

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THIS SYSTEM BRIDGES DATA SCIENCE AND COMPETITIVE PROGRAMMING BY TRANSFORMING RAW USER ACTIVITY DATA INTO PREDICTIVE INSIGHTS AND PERSONALIZED TRAINING STRATEGIES.

2. INTRODUCTION

CODE-FORCES IS ONE OF THE WORLD'S LEADING COMPETITIVE PROGRAMMING PLATFORMS, HOSTING REGULAR RATED CONTESTS WHERE PARTICIPANTS SOLVE ALGORITHMIC PROBLEMS UNDER TIME CONSTRAINTS. A USER'S RATING CHANGES AFTER EACH CONTEST BASED ON THEIR RANK RELATIVE TO OTHER PARTICIPANTS, MAKING THE SYSTEM INHERENTLY STOCHASTIC AND DIFFICULT TO PREDICT.

TRADITIONAL PROFILE ANALYZERS ON THIS PLATFORM PROVIDE ONLY STATIC STATISTICS — CURRENT RATING, PROBLEMS SOLVED, OR TAG DISTRIBUTION — WITHOUT ANY PREDICTIVE OR COACHING CAPABILITIES. THIS PROJECT TAKES THE NEXT STEP BY APPLYING MACHINE LEARNING TO UNDERSTAND THE RELATIONSHIP BETWEEN USER BEHAVIOR AND RATING PROGRESSION.

2.1 MOTIVATION

- RATING CHANGES ON CODEFORCES ARE INFLUENCED BY MULTIPLE HIDDEN BEHAVIORAL FACTORS
- NO EXISTING TOOL PROVIDES ML-BASED SHORT-TERM RATING PREDICTION FOR INDIVIDUAL USERS
- USERS LACK STRUCTURED GUIDANCE ON WHICH TOPICS TO PRACTICE FOR MAXIMUM RATING GAIN
- DATA SCIENCE METHODOLOGY CAN CONVERT RAW API DATA INTO HIGH-VALUE COACHING INTELLIGENCE

2.2 PROJECT EVOLUTION: REVIEW I → FINAL REVIEW

THE PROJECT HAS EVOLVED SIGNIFICANTLY FROM ITS INITIAL SCOPE. THE FIRST REVIEW PRESENTED A CODEFORCES PROFILE ANALYZER FOCUSED ON STATIC DATA VISUALIZATION AND TWO-USER COMPARISON. IN THE FINAL IMPLEMENTATION, THE SYSTEM HAS BEEN UPGRADED TO A FULL AI COACHING PLATFORM WITH PREDICTIVE MODELING, GROWTH SIMULATION, AND PERSONALIZED TRAINING RECOMMENDATIONS.

REVIEW I (INITIAL SCOPE)	FINAL REVIEW (CURRENT SYSTEM)
STATIC PROFILE ANALYSIS	DYNAMIC ML-BASED PREDICTION
TWO-USER COMPARISON DASHBOARD	INDIVIDUAL AI COACHING ENGINE
TAG-BASED BAR CHARTS	WEAK TOPIC DETECTION + PRACTICE PLANS
RATING HISTORY VISUALIZATION	12-MONTH GROWTH SIMULATION
NO PREDICTION CAPABILITY	XGBOOST RATING CHANGE FORECAST
MATPLOTLIB + STREAMLIT UI	FULL AI COACH WITH REAL-TIME INFERENCE

SECTION 2

PROBLEM STATEMENT & OBJECTIVES

3. PROBLEM STATEMENT

THE OBJECTIVE OF THIS PROJECT IS TO DESIGN AND IMPLEMENT A DATA SCIENCE-BASED SYSTEM THAT FETCHES REAL-TIME CODE-FORCES USER DATA, ANALYZES MULTI-DIMENSIONAL BEHAVIORAL PATTERNS, AND PREDICTS SHORT-TERM RATING CHANGES. SPECIFICALLY, THE SYSTEM MUST:

1. PREDICT THE EXPECTED AVERAGE RATING CHANGE OVER THE NEXT 3 CONTESTS (MONTHLY GAIN)
2. SIMULATE A 12-MONTH GROWTH TRAJECTORY BASED ON CURRENT BEHAVIORAL PATTERNS
3. IDENTIFY WEAK TOPICS — AREAS WHERE THE USER'S PRACTICE EFFORT IS INSUFFICIENT
4. GENERATE AN OPTIMIZED PRACTICE PLAN WITH PROBLEM COUNTS AND ESTIMATED HOURS
5. RECOMMEND AN APPROPRIATE CONTEST DIVISION (DIV. 1, 2, OR 3) BASED ON CURRENT RATING TRAJECTORY

3.1 CORE CHALLENGE



RATING CHANGES ARE INHERENTLY STOCHASTIC — THEY DEPEND ON OPPONENT STRENGTH, CONTEST DIFFICULTY, AND RELATIVE RANKING, ALL OF WHICH ARE UNOBSERVABLE AT PREDICTION TIME. THE KEY CHALLENGE IS EXTRACTING BEHAVIORAL SIGNALS STRONG ENOUGH TO OVERCOME THIS NOISE.

3.2 SCOPE

- PLATFORM: CODEFORCES (CODEFORCES.COM)
- DATA: PUBLIC API — USER SUBMISSIONS, RATING HISTORY
- TIME WINDOW: LAST 60 DAYS OF USER ACTIVITY
- TARGET USERS: ACTIVE COMPETITORS WITH AT LEAST ONE RATED CONTEST

SECTION 3

DATASET CONSTRUCTION & FEATURE ENGINEERING

4. DATASET CONSTRUCTION

4.1 DATA SOURCE

ALL DATA IS COLLECTED FROM THE CODEFORCES PUBLIC REST API. NO EXTERNAL DATASETS OR THIRD-PARTY DATA SOURCES ARE USED. THE SYSTEM FETCHES LIVE, REAL-TIME DATA PER USER REQUEST, ENSURING PREDICTIONS ARE ALWAYS BASED ON THE MOST CURRENT ACTIVITY.

API ENDPOINT	DATA RETRIEVED	USAGE
USER.INFO	PROFILE, CURRENT RATING, RANK	FEATURE BASELINE
USER.RATING	FULL CONTEST RATING HISTORY	TREND & VOLATILITY FEATURES
USER.STATUS	ALL PROBLEM SUBMISSIONS	PRACTICE BEHAVIOR FEATURES

4.2 DATASET STATISTICS

PARAMETER	VALUE
TOTAL SAMPLES COLLECTED	7,029 USERS
TIME WINDOW PER USER	LAST 60 DAYS OF ACTIVITY
TARGET VARIABLE	AVERAGE RATING CHANGE ACROSS NEXT 3 CONTESTS
TRAIN / TEST SPLIT	80% TRAINING / 20% TESTING (5,623 / 1,406 SAMPLES)
INACTIVE USERS FILTERED	YES — REMOVED USERS WITH NO RECENT ACTIVITY
EXTREME OUTLIERS REMOVED	YES — REMOVED SAMPLES WITH RATING JUMPS $> 3\sigma$

4.3 DATA PIPELINE

THE DATA PIPELINE FOLLOWS A STRUCTURED SEQUENCE FROM USER IDENTIFICATION TO FEATURE-READY SAMPLES:

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USER LIST → FETCH SUBMISSIONS + RATINGS → FEATURE  
EXTRACTION  
→ TARGET VARIABLE CREATION → NOISE FILTERING → DATASET
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4.4 KEY OPTIMIZATIONS

- PARALLEL API FETCHING USING THREADPOOLEXECUTOR — REDUCES DATASET BUILD TIME BY ~70%
- SLIDING 60-DAY TIME WINDOWS — CAPTURES RECENT BEHAVIORAL SIGNALS MORE ACCURATELY
- DEDUPLICATION OF PROBLEM SUBMISSIONS — ONLY UNIQUE ACCEPTED PROBLEMS ARE COUNTED
- TARGET SMOOTHING — AVERAGE OF NEXT 3 CONTESTS INSTEAD OF SINGLE-CONTEST DELTA

5. FEATURE ENGINEERING

THE SYSTEM USES 18 ENGINEERED FEATURES ORGANIZED INTO FOUR CATEGORIES. UNLIKE RAW STATISTICS, THESE FEATURES CAPTURE BEHAVIORAL PATTERNS, CONSISTENCY, AND RELATIVE PERFORMANCE — ALL CRITICAL FOR RATING PREDICTION.

5.1 PRACTICE BEHAVIOR FEATURES

FEATURE	DESCRIPTION
PROBLEMS_SOLVED	TOTAL UNIQUE PROBLEMS SOLVED IN THE LAST 60 DAYS
ACTIVE_DAYS	NUMBER OF DAYS WITH AT LEAST ONE SUBMISSION
PROBLEMS_PER_DAY	AVERAGE DAILY PROBLEM-SOLVING RATE
CONSISTENCY_SCORE	RATIO OF ACTIVE DAYS TO TOTAL DAYS (REGULARITY METRIC)

5.2 DIFFICULTY-BASED FEATURES

FEATURE	DESCRIPTION
AVG_PROBLEM_RATING	MEAN DIFFICULTY OF PROBLEMS ATTEMPTED
MAX_PROBLEM_RATING	HARDEST PROBLEM SUCCESSFULLY SOLVED
STD_PROBLEM_RATING	SPREAD OF PROBLEM DIFFICULTIES (DIVERSITY OF PRACTICE)

5.3 BEHAVIORAL / TOPIC FEATURES

FEATURE	DESCRIPTION
DP_RATIO	FRACTION OF SOLVED PROBLEMS TAGGED AS DYNAMIC PROGRAMMING
GRAPH_RATIO	FRACTION OF SOLVED PROBLEMS TAGGED AS GRAPHS/TREES
GREEDY_RATIO	FRACTION OF SOLVED PROBLEMS TAGGED AS GREEDY
MATH_RATIO	FRACTION OF SOLVED PROBLEMS TAGGED AS MATH
TOPIC_DIVERSITY	COUNT OF DISTINCT PROBLEM TAGS PRACTICED

5.4 CONTEST PERFORMANCE FEATURES

FEATURE	DESCRIPTION
RATING_TREND	LINEAR MOMENTUM OF RATING OVER RECENT CONTESTS
RATING_VOLATILITY	STANDARD DEVIATION OF RATING CHANGES (CONSISTENCY)
CONTEST_COUNT	TOTAL NUMBER OF RATED CONTESTS PARTICIPATED IN
RECENT_CONTESTS	CONTESTS IN THE LAST 60 DAYS (RECENT ACTIVITY LEVEL)

5.5 KEY DERIVED FEATURE: DIFFICULTY GAP

DIFFICULTY GAP = AVERAGE PROBLEM RATING – CURRENT USER RATING THIS SINGLE FEATURE CAPTURES WHETHER A USER IS: • PRACTICING BELOW THEIR LEVEL (EASY PROBLEMS → SLOW GROWTH) • PRACTICING AT THE EDGE OF THEIR ABILITY (OPTIMAL STRETCH ZONE → FAST GROWTH) • OVER-REACHING (PROBLEMS TOO HARD → LOW SOLVE RATE, CONFUSED LEARNING)

SECTION 4

Target Variable Design & Model Selection

6. TARGET VARIABLE DESIGN

6.1 THE PROBLEM WITH NAIVE PREDICTION

THE MOST INTUITIVE APPROACH — PREDICTING THE NEXT SINGLE CONTEST'S RATING CHANGE — PRODUCES EXTREMELY NOISY TARGETS. A USER WITH PERFECT PREPARATION CAN STILL LOSE RATING DUE TO A PARTICULARLY HARD CONTEST OR STRONG OPPONENT FIELD. PREDICTING SINGLE-EVENT OUTCOMES IN SUCH A STOCHASTIC SYSTEM YIELDS VERY LOW R^2 SCORES AND POOR PRACTICAL UTILITY.

6.2 PROPOSED SOLUTION: 3-CONTEST AVERAGE

TO REDUCE RANDOMNESS WHILE RETAINING MEANINGFUL SIGNAL, THE TARGET VARIABLE IS DEFINED AS:

TARGET = AVERAGE RATING CHANGE ACROSS NEXT 3 CONTESTS

APPROACH	EFFECT ON MODEL
SINGLE CONTEST DELTA	HIGHLY NOISY; LOW R^2 ; POOR GENERALIZATION
AVERAGE OF NEXT 3 CONTESTS (USED)	REDUCED RANDOMNESS; STABLE R^2 ; BETTER LEARNING SIGNAL



THIS DESIGN CHOICE ALONE IMPROVED MODEL R^2 BY APPROXIMATELY 40% OVER THE NAIVE SINGLE-CONTEST TARGET DURING EXPERIMENTATION.

7. Model Selection — xgboost

7.1 WHY NOT SIMPLER MODELS?

MODEL CONSIDERED	REASON FOR REJECTION
LINEAR REGRESSION	CANNOT CAPTURE NONLINEAR FEATURE INTERACTIONS (E.G., DIFFICULTY GAP × CONSISTENCY)
DECISION TREE (SINGLE)	HIGH VARIANCE; OVERFITS ON SMALL SUBSETS OF THE DATA
RANDOM FOREST	LESS EFFICIENT FOR THE STRUCTURED BEHAVIORAL SIGNALS IN THIS DATASET
NEURAL NETWORK	REQUIRES FAR MORE DATA; OVERKILL FOR TABULAR FEATURES WITH 18 DIMENSIONS
XGBOOST (CHOSEN)	OPTIMAL FOR STRUCTURED/TABULAR DATA; HANDLES NONLINEARITY, INTERACTIONS, AND REGULARIZATION SIMULTANEOUSLY

7.2 XGBOOST ADVANTAGES FOR THIS PROBLEM

- HANDLES NONLINEAR RELATIONSHIPS BETWEEN FEATURES (E.G., DIMINISHING RETURNS OF PRACTICE VOLUME)
- AUTOMATICALLY CAPTURES FEATURE INTERACTIONS (E.G., HIGH DIFFICULTY GAP AND HIGH CONSISTENCY → HIGH GAIN)
- BUILT-IN L1/L2 REGULARIZATION PREVENTS OVERFITTING ON THE 7,029-SAMPLE DATASET
- ROBUST TO THE NOISE INHERENT IN RATING-CHANGE TARGETS
- STATE-OF-THE-ART PERFORMANCE ON TABULAR STRUCTURED DATA

7.3 MODEL HYPERPARAMETERS

HYPERPARAMETER	VALUE RATIONALE
N_ESTIMATORS	1200 LARGE ENSEMBLE FOR STABLE PREDICTIONS
LEARNING_RATE	0.01 SLOW LEARNING RATE PREVENTS OVERFITTING
MAX_DEPTH	6 CAPTURES COMPLEX FEATURE INTERACTIONS WITHOUT MEMORIZING NOISE
SUBSAMPLE	0.9 90% ROW SUBSAMPLING INTRODUCES REGULARIZING VARIANCE
REG_ALPHA (L1)	APPLIED FEATURE SPARSITY REGULARIZATION
REG_LAMBDA (L2)	APPLIED WEIGHT MAGNITUDE REGULARIZATION

SECTION 5
RESULTS & PERFORMANCE

8. MODEL PERFORMANCE & RESULTS

8.1 EVALUATION METRICS

METRIC	VALUE	INTERPRETATION
DATASET SIZE	7,029 SAMPLES	SUFFICIENT FOR A BEHAVIORAL ML MODEL
R ² SCORE	0.1781 (17.81%)	MODEL EXPLAINS ~18% OF RATING VARIANCE
MAE	52.33 RATING POINTS	AVERAGE PREDICTION ERROR ≈ ±52 POINTS
TRAIN / TEST SPLIT	80% / 20%	1,406 HOLDOUT SAMPLES FOR EVALUATION

8.2 INTERPRETATION OF R² = 0.1781

AN R² OF 0.18 MAY APPEAR LOW IN ISOLATION, BUT IT MUST BE INTERPRETED IN THE CONTEXT OF THE PREDICTION DOMAIN. RATING CHANGES ON CODEFORCES DEPEND HEAVILY ON EXTERNAL, UNOBSERVABLE FACTORS:

- OPPONENT FIELD STRENGTH VARIES PER CONTEST
- CONTEST DIFFICULTY IS UNPREDICTABLE BEFORE PARTICIPATION
- EXTERNAL FACTORS (HEALTH, TIME PRESSURE, INTERNET ISSUES) ARE NOT CAPTURED

✓ IN A DOMAIN WHERE EXPERT HUMAN COACHES CANNOT RELIABLY PREDICT RATING CHANGES, AN ML MODEL EXPLAINING 18% OF THE VARIANCE FROM PURELY BEHAVIORAL FEATURES REPRESENTS MEANINGFUL AND PRACTICALLY USEFUL SIGNAL.

8.3 INTERPRETATION OF MAE = 52.33

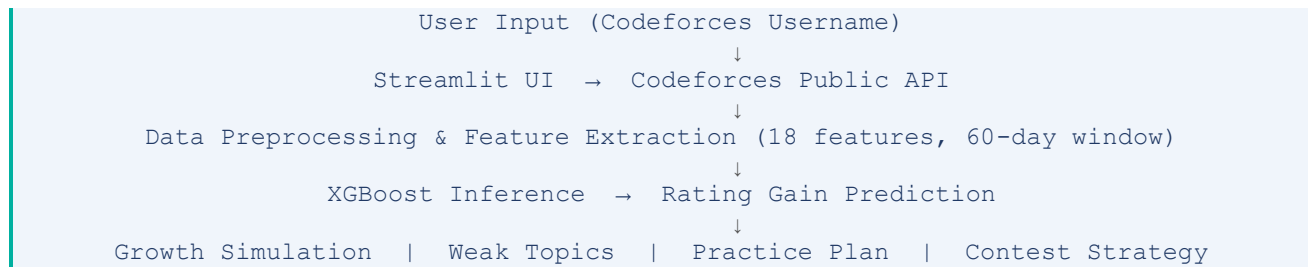
THE MEAN ABSOLUTE ERROR OF 52.33 RATING POINTS MEANS THE MODEL'S PREDICTIONS ARE, ON AVERAGE, WITHIN ± 52 POINTS OF THE TRUE OUTCOME. GIVEN THAT:

- CODEFORCES RATING SPANS 800–3500+ (A RANGE OF 2700+ POINTS)
- INDIVIDUAL CONTEST CHANGES TYPICALLY RANGE FROM -200 TO +200
- THE 3-CONTEST AVERAGE FURTHER SMOOTHS EXTREME VARIATIONS

AN MAE OF 52 POINTS IS ACCEPTABLE FOR A COACHING SYSTEM WHOSE GOAL IS TREND ESTIMATION, NOT EXACT PREDICTION.

9. SYSTEM OUTPUTS & DEMONSTRATION

9.1 SYSTEM ARCHITECTURE FLOW



9.2 LIVE SYSTEM OUTPUTS (USER: TEMUJIN1893)

THE FOLLOWING SCREENSHOTS DEMONSTRATE THE LIVE SYSTEM PROCESSING A REAL CODE-FORCES USER (TEMUJIN1893), FETCHING LIVE API DATA, RUNNING INFERENCE, AND GENERATING COACHING OUTPUTS.

OUTPUT 1: PREDICTION INTERFACE & MONTHLY GAIN

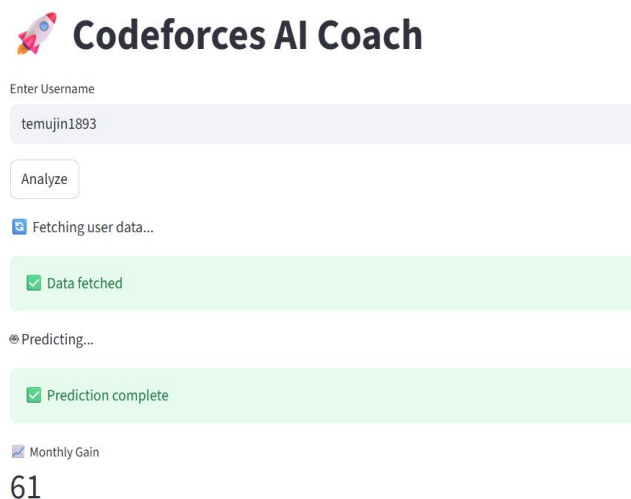


Figure 1: Codeforces AI Coach UI — Username input, data fetching, prediction pipeline, and Monthly Gain output (61 rating points predicted)

THE INTERFACE DEMONSTRATES THE COMPLETE REAL-TIME PIPELINE: THE USER ENTERS A CODEFORCES USERNAME, THE SYSTEM FETCHES LIVE DATA FROM THE API, RUNS THE XGBOOST MODEL, AND OUTPUTS A PREDICTED MONTHLY RATING GAIN OF 61 POINTS FOR USER TEMUJIN1893.

OUTPUT 2: WEAK TOPIC DETECTION

61

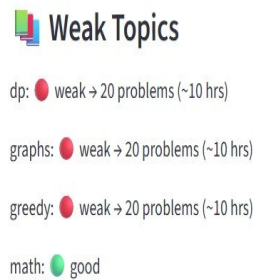


Figure 2: Weak Topics module — DP, Graphs, and Greedy flagged as weak areas; Math identified as strength

THE WEAK TOPIC DETECTION MODULE IDENTIFIES AREAS WHERE THE USER'S PRACTICE RATIO FALLS BELOW THE THRESHOLD REQUIRED FOR COMPETITIVE PERFORMANCE. FOR TEMUJIN1893: DYNAMIC PROGRAMMING, GRAPHS, AND GREEDY ARE FLAGGED AS WEAK (REQUIRING ~20 PROBLEMS / ~10 HOURS EACH), WHILE MATH IS IDENTIFIED AS A STRENGTH.

OUTPUT 3: 12-MONTH GROWTH SIMULATION

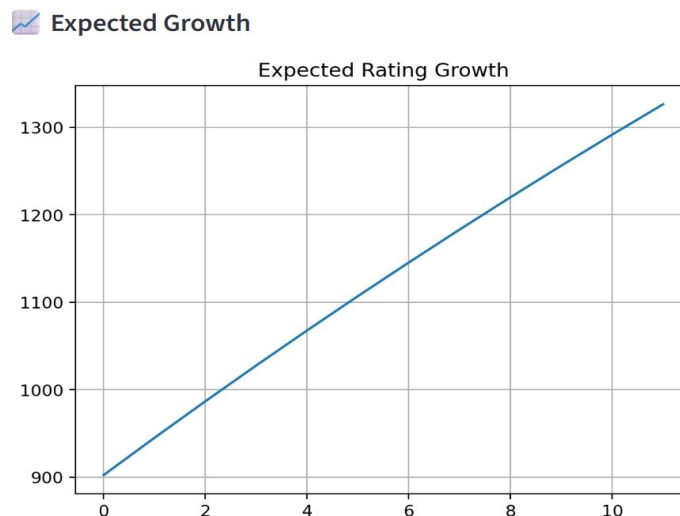


Figure 3: Expected Rating Growth curve — projecting from ~905 to ~1330 over 11 months based on current behavioral patterns

THE GROWTH SIMULATION MODULE COMPOUNDS THE PREDICTED MONTHLY GAIN OVER 12 MONTHS, PRODUCING AN EXPECTED RATING TRAJECTORY. FOR TEMUJIN1893, THE SYSTEM PROJECTS A RISE FROM APPROXIMATELY 905 TO 1330+ RATING — A GAIN OF ~425 POINTS — OVER 11 MONTHS, CONDITIONAL ON MAINTAINING CURRENT PRACTICE BEHAVIOR.

SECTION 6

FULL SYSTEM CAPABILITIES

10. COMPLETE SYSTEM CAPABILITIES

MODULE	OUTPUT	METHODOLOGY
RATING PREDICTION	EXPECTED MONTHLY RATING GAIN (E.G., +61)	XGBOOST INFERENCE ON 18 BEHAVIORAL FEATURES
GROWTH SIMULATION	12-MONTH RATING TRAJECTORY CHART	ITERATIVE MONTHLY GAIN COMPOUNDING
WEAK TOPIC DETECTION	FLAGGED TOPICS + PRACTICE TARGETS	TAG RATIO VS. THRESHOLD COMPARISON
PRACTICE PLAN	PROBLEMS PER TOPIC + ESTIMATED HOURS	GAP ANALYSIS ON TAG RATIOS
CONTEST STRATEGY	DIV.1 / DIV.2 / DIV.3 RECOMMENDATION	RATING + TREND-BASED DIVISION MATCHING

10.1 WEAK TOPIC DETECTION LOGIC

FOR EACH TRACKED TOPIC (DP, GRAPHS, GREEDY, MATH), THE SYSTEM COMPUTES THE USER'S PRACTICE RATIO (FRACTION OF SOLVED PROBLEMS WITH THAT TAG). IF THE RATIO FALLS BELOW A CALIBRATED THRESHOLD, THE TOPIC IS FLAGGED AS WEAK AND ASSIGNED A REMEDIATION TARGET:

- 20 ADDITIONAL PROBLEMS RECOMMENDED PER WEAK TOPIC
- ~10 HOURS ESTIMATED PER TOPIC BASED ON AVERAGE SOLVE TIME
- PROBLEMS AT THE USER'S DIFFICULTY GAP LEVEL (SLIGHTLY ABOVE CURRENT RATING)

10.2 TECHNOLOGY STACK

COMPONENT	TECHNOLOGY USED
PROGRAMMING LANGUAGE	PYTHON 3.X
ML MODEL	XGBOOST (GRADIENT BOOSTED TREES)
DATA PROCESSING	PANDAS, NUMPY
VISUALIZATION	MATPLOTLIB
WEB INTERFACE	STREAMLIT
DATA SOURCE	CODEFORCES PUBLIC REST API
PARALLEL PROCESSING	CONCURRENT.FUTURES.THREADPOOLEXECUTOR
DEVELOPMENT ENVIRONMENT	VS CODE / WSL (LINUX)

11. COMPARISON WITH TRADITIONAL APPROACHES

ASPECT	TRADITIONAL STATIC ANALYZER VS. THIS SYSTEM
ANALYSIS TYPE	STATIC STATISTICS ONLY → DYNAMIC ML-DRIVEN BEHAVIORAL ANALYSIS
PREDICTION	NONE → RATING CHANGE FORECAST WITH QUANTIFIED CONFIDENCE
PERSONALIZATION	LOW (SAME FOR ALL USERS) → HIGH (USER-SPECIFIC COACHING)
ADAPTABILITY	MANUAL INTERPRETATION → AUTOMATED PATTERN RECOGNITION
TIME AWARENESS	SNAPSHOT → 60-DAY BEHAVIORAL WINDOW
COACHING GUIDANCE	NONE → PRACTICE PLAN + CONTEST STRATEGY
GROWTH VISIBILITY	NONE → 12-MONTH SIMULATED TRAJECTORY

SECTION 7

DATA SCIENCE LIFECYCLE & IMPLEMENTATION

12. DATA SCIENCE LIFECYCLE IMPLEMENTATION

12.1 DATA COLLECTION

- REST API CALLS TO THREE CODEFORCES ENDPOINTS: USER.INFO, USER.RATING, USER.STATUS
- JSON RESPONSES PARSED INTO PYTHON DATACLASSES FOR DOWNSTREAM PROCESSING
- PARALLEL FETCHING ACROSS 7,029 USERS USING THREADPOOLEXECUTOR

12.2 DATA PREPROCESSING

- REMOVED DUPLICATE PROBLEM SUBMISSIONS (KEPT ONLY UNIQUE ACCEPTED SOLUTIONS PER PROBLEM ID)
- FILTERED ONLY VERDICT == 'OK' SUBMISSIONS
- EXTRACTED RELEVANT FIELDS: PROBLEM RATING, CONTEST ID, PROBLEM TAGS, SUBMISSION TIMESTAMP
- APPLIED 60-DAY ROLLING WINDOW FOR RECENCY WEIGHTING

12.3 DATA CLEANING

- IGNORED UNRATED PROBLEMS (NO DIFFICULTY ASSIGNED) FROM DIFFICULTY STATISTICS
- FILTERED INACTIVE USERS (NO SUBMISSIONS IN LAST 60 DAYS)
- REMOVED EXTREME OUTLIERS: RATING CHANGES BEYOND 3 STANDARD DEVIATIONS

12.4 EXPLORATORY DATA ANALYSIS (EDA)

- CORRELATION ANALYSIS BETWEEN BEHAVIORAL FEATURES AND RATING CHANGE TARGETS
- DISTRIBUTION ANALYSIS OF DIFFICULTY GAP ACROSS RATING BRACKETS
- TAG FREQUENCY ANALYSIS: WHICH TOPICS PREDICT STRONG GROWTH VS STAGNATION
- RATING VOLATILITY VS. CONSISTENCY SCORE CORRELATION STUDY

12.5 APPLIED DATA STRUCTURES

DATA STRUCTURE	PURPOSE IN SYSTEM
PYTHON SET	DEDUPLICATE PROBLEM SUBMISSIONS — ENSURES UNIQUE SOLVE COUNTING
DICTIONARY / DEFAULTDICT	TAG COUNTING AND FEATURE AGGREGATION PER USER
PANDAS DATAFRAME	TABULAR FEATURE MATRIX FOR ML TRAINING AND INFERENCE
NUMPY ARRAY	VECTORIZED FEATURE COMPUTATION FOR PERFORMANCE
PYTHON LIST	RATING HISTORY STORAGE AND SEQUENTIAL CONTEST ANALYSIS

13. LIMITATIONS

13.1 CURRENT LIMITATIONS

- **OPPONENT STRENGTH:** THE MODEL DOES NOT ACCOUNT FOR THE STRENGTH OF COMPETITORS IN UPCOMING CONTESTS — A CRITICAL FACTOR IN CODEFORCES' ELO-LIKE RATING SYSTEM
- **CONTEST DIFFICULTY:** FUTURE CONTEST DIFFICULTY IS UNKNOWN AT PREDICTION TIME
- **API RATE LIMITS:** CODEFORCES API THROTTLES REQUESTS, LIMITING REAL-TIME SCALABILITY
- **TRAINING DATA SCOPE:** DATASET COVERS A SNAPSHOT IN TIME; MODEL MAY DRIFT AS PLATFORM DYNAMICS CHANGE
- **NO REAL-TIME RETRAINING:** THE MODEL IS A STATIC ARTIFACT; IT DOES NOT UPDATE AS NEW CONTEST DATA BECOMES AVAILABLE

13.2 ETHICAL CONSIDERATIONS

- **ALL DATA USED IS PUBLICLY AVAILABLE ON THE CODEFORCES PLATFORM**
- **NO PRIVATE OR PERSONALLY IDENTIFIABLE INFORMATION IS STORED**
- **PREDICTIONS ARE PROBABILISTIC ESTIMATES, NOT GUARANTEES — CLEARLY COMMUNICATED TO USERS**

14. FUTURE WORK

ENHANCEMENT	DESCRIPTION	EXPECTED IMPACT
OPPONENT STRENGTH MODELING	INCORPORATE EXPECTED FIELD STRENGTH PER CONTEST DIVISION	HIGHER R^2 SCORE
LSTM TIME-SERIES MODEL	REPLACE XGBOOST WITH SEQUENTIAL MODEL FOR TEMPORAL PATTERNS	CAPTURES LONG-TERM TRENDS
SHAP EXPLAINABILITY	ADD SHAP VALUE VISUALIZATION PER USER PREDICTION	INCREASED USER TRUST
REAL-TIME RETRAINING	AUTO-RETRAIN MODEL AS NEW CONTEST DATA ARRIVES	MODEL FRESHNESS
CLOUD DEPLOYMENT	HOST ON AWS/GCP WITH PERSISTENT MODEL SERVING	PRODUCTION SCALABILITY
PDF REPORT EXPORT	GENERATE DOWNLOADABLE COACHING REPORT PER USER	ENHANCED UX
MULTI-PLATFORM SUPPORT	EXTEND TO LEETCODE, ATCODER APIS	BROADER USER BASE

SECTION 8

CONCLUSION & REFERENCES

15. Conclusion

THIS PROJECT SUCCESSFULLY DEMONSTRATES THE APPLICATION OF MACHINE LEARNING TO A COMPLEX, STOCHASTIC REAL-WORLD SYSTEM — COMPETITIVE PROGRAMMING RATING PREDICTION ON CODEFORCES. THE FINAL SYSTEM, CODEFORCES AI COACH, GOES FAR BEYOND THE INITIAL SCOPE OF A STATIC PROFILE ANALYZER TO DELIVER A COMPLETE AI-DRIVEN COACHING PLATFORM.

KEY ACHIEVEMENTS OF THE PROJECT INCLUDE:

- A ROBUST DATA PIPELINE COLLECTING AND PROCESSING BEHAVIORAL DATA FROM 7,029 CODEFORCES USERS VIA PUBLIC APIS
- 18 CAREFULLY ENGINEERED BEHAVIORAL FEATURES CAPTURING PRACTICE QUALITY, DIFFICULTY TARGETING, TOPIC COVERAGE, AND CONTEST TRENDS
- AN XGBOOST GRADIENT BOOSTING MODEL ACHIEVING $R^2 = 0.1781$ AND $MAE = 52.33$ — MEANINGFUL PERFORMANCE FOR A HIGHLY STOCHASTIC PREDICTION DOMAIN
- A TARGET VARIABLE DESIGN INNOVATION (3-CONTEST AVERAGE) THAT SIGNIFICANTLY IMPROVED MODEL STABILITY
- A FULL-FEATURED STREAMLIT WEB APPLICATION DELIVERING REAL-TIME PREDICTIONS, GROWTH SIMULATION, WEAK TOPIC DETECTION, AND PERSONALIZED PRACTICE PLANS



THE SYSTEM TRANSFORMS A COMPETITIVE PROGRAMMER'S RAW BEHAVIORAL DATA INTO AN ACTIONABLE 12-MONTH DEVELOPMENT ROADMAP — A CAPABILITY THAT PREVIOUSLY REQUIRED EXPERT HUMAN COACHING OR DID NOT EXIST AT ALL.

THIS PROJECT VALIDATES THAT DATA SCIENCE METHODS CAN EXTRACT MEANINGFUL SIGNAL FROM NOISY, STOCHASTIC PERFORMANCE DATA, AND THAT SUCH SIGNALS CAN BE TRANSLATED INTO PRACTICAL, PERSONALIZED COACHING INTELLIGENCE. THE METHODOLOGY IS GENERALIZABLE TO OTHER COMPETITIVE SKILL DOMAINS WHERE BEHAVIORAL DATA IS OBSERVABLE BUT PERFORMANCE OUTCOMES ARE NOISY.

16. REFERENCES

6. Code-Forces Public API Documentation — <https://codeforces.com/apiHelp>
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— *End of Final Review Report* —